

Optimal Adjuvant Chemotherapy and Targeted Therapy for Early Breast Cancer—Cyclin-Dependent Kinase 4 and 6 Inhibitors: ASCO Guideline Rapid Recommendation Update

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ASCO Rapid Recommendation Updates highlight revisions to select ASCO guideline recommendations as a response to the emergence of new and practice-changing data. The rapid updates are supported by an evidence review and follow the guideline development processes outlined in the [ASCO Guideline Methodology Manual](#). The goal of these articles is to disseminate updated recommendations, in a timely manner, to better inform health practitioners and the public on the best available cancer care options. Guidelines and updates are not intended to substitute for independent professional judgment of the treating provider and do not account for individual variation among patients. See appendix for disclaimers and other important information ([Appendix 1](#) and [Appendix 2](#), online only).

ACCOMPANYING CONTENT

 Listen to the podcast by Dr Freedman and Dr Giordano at <https://ascopubs.org/journal/jco/asco-guidelines-podcast>

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BACKGROUND

In 2020, ASCO published a guideline on the selection of optimal adjuvant chemotherapy and targeted therapy for breast cancer.¹ This is a rapid recommendation update of that guideline that addresses, in turn, the adjuvant use of the cyclin-dependent kinase 4 and 6 (CDK4/6) inhibitors abemaciclib and ribociclib in patients with stage II and III breast cancer.

Abemaciclib With Endocrine Therapy in the Treatment of High-Risk Early Breast Cancer

A 2022 update occasioned by the publication of monarchE² recommended 2 years of abemaciclib (150 mg twice daily) plus endocrine therapy (ET) for patients with node-positive, hormone receptor-positive human epidermal growth factor receptor 2 (HER2)-negative breast cancer at high risk of recurrence and a Ki-67 score of $\geq 20\%$ as determined by a US Food and Drug Administration (FDA)-approved test.³ The FDA expanded the approval of abemaciclib on March 3, 2023; this expanded approval eliminates the Ki-67 testing requirement, acknowledging the benefits observed across the intention-to-treat (ITT) population of the monarchE study.⁴ Recently published data⁵ at a median follow-up of 54 months demonstrate a sustained hazard ratio (HR) of 0.680 (95% CI, 0.599 to 0.772) for invasive disease-free survival (IDFS) for abemaciclib plus ET, with a deepening of benefit over time and 5-year absolute improvement in IDFS of 7.6% and distant relapse-free survival of 6.7% compared with ET alone. Although fewer deaths have been reported in the abemaciclib-treated arm, statistical significance has not been reached.

Ribociclib With ET in the Treatment of Early Breast Cancer

Slamon et al⁶ recently published the first results of a prespecified interim analysis of data from NATALEE, an open-label, multicenter, randomized phase III trial that evaluated the addition of 3 years of ribociclib (400 mg once daily, 3 weeks on followed by 1 week off) to standard ET (letrozole 2.5 mg once per day or anastrozole 1 mg once per day for ≥ 5 years) versus ET alone among women and men with stage II to III, hormone receptor-positive, HER2-negative early breast cancer (EBC). Premenopausal women and men received gonadal suppression as part of their ET. For those with stage IIA disease (T2No), at least one of the following was also required: grade 2



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disease plus Ki-67 $\geq 20\%$ or high genomic risk or grade 3 disease. Significant improvements in the primary study end point of IDFS constituted a signal for further amending the 2020 ASCO guideline.

METHODS

A literature search was conducted to identify phase III trials on the use of ribociclib or abemaciclib in this patient population. No additional trials were identified.

EVIDENCE REVIEW FOR ADJUVANT RIBOCICLIB

NATALEE randomly assigned patients (1:1) to ribociclib 400 mg plus ET ($n = 2,549$) or ET alone ($n = 2,552$). At a median follow-up of 27.7 months, adjuvant ribociclib plus ET significantly improved IDFS compared with ET alone (HR, 0.75 [95% CI, 0.62 to 0.91]; $P = .003$). The 3-year IDFS rates were 90.4% versus 87.1%, with a 3.3% absolute IDFS benefit at 3 years with ribociclib-based therapy. Ribociclib plus ET significantly improved distant disease-free survival (DDFS) compared with ET alone (HR, 0.74; 95% CI, 0.60 to 0.91); 3-year DDFS rates were 90.8% versus 88.6%. At a median follow-up of 30 months, overall survival was not significantly different between treatment arms (HR, 0.76; 95% CI, 0.54 to 1.07), with longer-term follow-up planned.

The following grade 3 or higher adverse events (AEs) of special interest occurred in $\geq 15\%$ of patients in the ribociclib plus ET arm and were numerically higher in the ribociclib plus ET arm compared with the ET arm: neutropenia (43.8% v 0.8%) and liver-related AEs (8.3% v 1.5%). Liver-related AEs were the most frequent AE of any grade leading to ribociclib discontinuation (8.9%). Any-grade QT prolongation occurred in 5.2% of those receiving ribociclib plus ET and 1.2% for those receiving ET alone. For AEs of any grade, nausea (23% v 7.5%), headache (22% v 16.5%), and fatigue (21.9% v 12.7%) were numerically higher in the ribociclib plus ET arm compared with the ET arm. Appraisal of the trial report using the GRADE instrument⁷ was performed as per ASCO's methodology and found a high certainty of the evidence.

UPDATED RECOMMENDATIONS

Recommendation 1

Abemaciclib for 2 years plus ET for ≥ 5 years may be offered to patients meeting the criteria of the ITT monarchE population with resected, hormone receptor-positive, HER2-negative, node-positive, early breast cancer at high risk of recurrence, defined as having ≥ 4 positive axillary lymph nodes (ALNs) or as having 1–3 positive ALNs plus at least one of the following features: grade 3 disease, tumor size ≥ 5 cm, or Ki-67 index $\geq 20\%$. Although the FDA's language is broad,

the Panel promotes the use of abemaciclib primarily in those who would have been eligible for monarchE based on that trial's eligibility criteria (Evidence quality: High; Strength of recommendation: Strong).

Recommendation 2

The Panel recommends, based on the phase III NATALEE trial, that adjuvant ribociclib (400 mg once daily, 3 weeks on followed by 1 week off) for 3 years plus ET may be offered to patients with anatomic stage II or III breast cancer who would have met criteria for study entry and have a high risk of recurrence (Evidence quality: High; Strength of recommendation: Conditional).

Qualifying Statements for Recommendations 1 and 2 on the Use of Adjuvant Abemaciclib and Ribociclib

The Panel believes that adjuvant CDK4/6 inhibitor therapy may not provide meaningful clinical benefit to all patients who would have been eligible for the available trials, especially the lower-risk patients who were included in the NATALEE trial. For example, for most patients with node-negative disease, the risks of ribociclib may outweigh the benefits, with the exception of some patients with the highest risk, node-negative disease. However, the Panel acknowledges that there are insufficient data to specify which subgroups of patients do or do not warrant therapy. The Panel thus recommends considering the benefits, risks, costs, and preferences for each individual patient when deciding whether to recommend therapy.

Among patients meeting criteria for both monarchE and NATALEE, the Panel also notes that, of the two CDK4/6 inhibitors, abemaciclib has longer follow-up, a deepening benefit over time, a shorter duration of therapy, and FDA approval in the adjuvant setting. In this case, the Panel favors using abemaciclib, reserving use of ribociclib in patients who have a contraindication to (eg, preexisting high-grade diarrhea) or intolerance of abemaciclib. The Panel characterized the strength of the ribociclib recommendation as conditional, pending future efficacy data and regulatory updates. Although a formal cost-effectiveness analysis was out of scope for this update, it could be informative for some decision makers considering the costs of both medications. Results from longer-term follow-up will further inform adjuvant therapy decision making.

Evidence supporting unchanged recommendations and sections on cost implications and more are found in the full guideline publication, and apply to this Rapid Update.¹ Additionally, for guideline tools and resources, including a complete summary table, visit www.asco.org/breast-cancer-guidelines.

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R.A.F. and S.H.G. were the Expert Panel Co-Chairs.

EDITOR'S NOTE

This ASCO Clinical Practice Guideline Recommendation Update provides a recommendation update, with review and analysis of the relevant literature for the recommendation. Additional information, including links to patient information at www.cancer.net, is available at www.asco.org/breast-cancer-guidelines.

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

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AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Optimal Adjuvant Chemotherapy and Targeted Therapy for Early Breast Cancer—Cyclin-Dependent Kinase 4 and 6 Inhibitors: ASCO Guideline Rapid Recommendation Update**

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APPENDIX 3. OPTIMAL ADJUVANT CHEMOTHERAPY AND TARGETED THERAPY FOR EARLY BREAST CANCER GUIDELINE EXPERT PANEL

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